

1. Questions

Study the following information carefully and answer the questions given below:

Ten persons - A, B, C, D, E, F, G, H, I and J are attending the seminar on two different dates, either 18th or 23rd of five different months, viz., January, April, July, September and December of the same year. Only one person attends the seminar on each date and only two persons attend the seminar in each month.

Condition:

I). The person's name starts with a vowel, doesn't attend the seminar in July.

II). The person's name starts with a vowel, attend the seminar on an odd-numbered date in reverse order as per the English alphabetical order from January to December.

J attends the seminar on an even numbered date in the month, having an odd number of days. Only two persons attend the seminar between G and the one who attends the seminar immediately after J. G and I are adjacent persons, but attend the seminar in different months. As many persons attend the seminar before I as after D. The one who attends the seminar adjacent to D attends the seminar five months after F. Only two months between F and B. The number of persons attending the seminar between B and A is **one less than** the number of persons attending the seminar between E and H. At most one person attends the seminar between C and D.

Which of the following month will the meaningful word “BAG” come based on the final arrangement?

- a. April and July
- b. December and January
- c. September and April
- d. April and December
- e. July and January

2. Questions

The number of persons attending the seminar between H and J is ____ the number of persons attending the seminar between E and F?

- a. One more than
- b. Two less than
- c. Same as
- d. One less than
- e. Two more than

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3. Questions

Which of the following statement(s) is/are TRUE with respect to the final arrangement?

- a. Only one person attends the seminar between D and J
- b. C doesn't attend the seminar in the month having an even number of days
- c. G and F attend the seminar on different dates
- d. D and A attend the seminar on the same date
- e. I attended the seminar immediately before H

4. Questions

If all persons are arranged in the alphabetical order from December to January, then the position of how many persons remains unchanged?

- a. Two
- b. Three
- c. Five
- d. One
- e. Four

5. Questions

If D is related to C, J is related to G in a certain way then in the same way, who among the following person is related to F?

- a. The one who attends the seminar on an even numbered date
- b. E
- c. The one who attends the seminar in July
- d. H
- e. The one who attends the seminar immediately after F

6. Questions

Study the following information carefully and answer the below questions:

Six triangles numbered 1 to 6 are inscribed in a hexagonal box.

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Each triangle has a letter and a value.

The value for each triangle is the difference between the square value of the first six odd numbers (except 1) and the first six numbers divisible by three. Then, the resultant value is multiplied with the first six natural numbers and then the values are arranged in order from triangles 1 to 6, respectively.

The letter for the triangle is the corresponding letter of the place value, which is the sum of the digits of the value of each triangle, thus obtained as per the English alphabetical series. Then each obtained letter is changed into its second preceding letter. Then the letters are arranged from triangles 1 to 6, respectively.

Conditions:

- I).** If the triangle has a consonant letter, then the letter is replaced by the second succeeding letter as per the English alphabetical series.
- II).** If the triangle has values divisible by 4, then the difference between the sum of the odd digits and the sum of the even digits is taken.
- III).** If the triangle has a vowel letter, then the letter is replaced by the third preceding letter as per the English alphabetical series.
- IV).** If the triangle has a value not divisible by 4, then the value is the square of the sum of the digits of the number.

What is the sum of the numbers of triangles 1, 2 and 4 based on the final arrangement?

- a. 272
- b. 165
- c. 154
- d. 214
- e. 158

7. Questions

How many triangles have a vowel letter with respect to the final output?

- a. One
- b. Two
- c. None
- d. Three
- e. Five

8. Questions

What is the final output of triangle 1 and triangle 3, respectively?

- a. 64H and 1P
- b. 36F and 225P
- c. 1X and 121F
- d. 36F and 1X
- e. 121F and 1P

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9. Questions

What is the difference between the place value (as per the English alphabetical order) of the letters which are obtained in the final output of triangle 3 and triangle 1 together and triangle 2?

- a. 12
- b. 10
- c. 16
- d. 24
- e. 18

10. Questions

If the output triangle values are replaced with their square values and the letters are replaced by the fifth preceding letter as per the English alphabetical order. Then, what could be the new values and letters of triangle 1 and 6, respectively?

- a. 6J and 18I
- b. 13M and 4R
- c. 6A and 15J
- d. 12L and 7H
- e. 5N and 14D

11. Questions

Study the following information carefully and answer the questions given below:

“A#B(12)” means “A is 15m to the north of B”.

“A@B(17)” means “A is 14m to the south of B”.

“A&B(21)” means “A is 23m west of B”.

“A^B(24)” means “A is 20m east of B”.

Condition:

K@L(15), C#E(12), F@J(17), D@H(11), E&L(8), H^C(16), D&F(4), J^A(9)

What is the position of H with respect to K based on the final arrangement?

- a. North west
- b. South west
- c. North east
- d. South east
- e. South

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12. Questions

What is the shortest distance between F and C?

- a. 18m
- b. $\sqrt{363}$ m
- c. 24m
- d. $\sqrt{388}$ m
- e. 22m

13. Questions

Four of the following five are alike in a certain way based on the given arrangement and thus form a group. Which of the following does not belong to that group?

- a. JD
- b. EH
- c. FK
- d. AE
- e. HL

14. Questions

If B@D(19), then what is the total distance between B and K?

- a. 5m
- b. 6m
- c. 7m
- d. 8m
- e. 10m

15. Questions

If X is the midpoint of C and H, the distance between the X and C is equal to the distance between ___ and ___?

- a. E and L
- b. F and J
- c. A and J
- d. K and E
- e. D and F

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16. Questions

Study the following information carefully and answer the questions given below:

Seven persons – P, Q, R, S, T, U and V have a different age and earn a different salary.

Note-I: The age of the husband is more than the age of his wife.

Note-II: If they all earn, their salaries will be multiples of a thousand and the salary of each person lies between 15k to 55k unless stated.

R is the sister-in-law of Q and is younger than T. Q earns 37k and is 27 years older than U. S is the son-in-law of P and older than R. V earns 7k more than S. T is the brother-in-law of S, who doesn't have any siblings. T is 5 years older than Q. V is the father in-law of R and earns 11k more than Q. T is the maternal uncle of U and younger than S. P earns 4k more than V and is older than S. R earns 23k less than P and is younger than Q and is older than U, who is the granddaughter of V. U's age is 15 years.

How is S related to U's uncle based on the final arrangement?

- a. Mother
- b. Father
- c. Brother-in-law
- d. Sister-in-law
- e. Brother

17. Questions

What is the total salary of V and R as per the final arrangement?

- a. 72,000
- b. 77,000
- c. 65,000
- d. 69,000
- e. 78,000

18. Questions

How many persons are older than Q based on the final arrangement?

- a. One
- b. Two
- c. Three
- d. Four
- e. More than four

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19. Questions

If P is 18 years older than T and 9 years younger than V, then what is the sum of the age of V and P?

- a. 139 years
- b. 123 years
- c. 135 years
- d. 132 years
- e. 147 years

20. Questions

Which of the following statement(s) is /are TRUE with respect to the final arrangement?

- a. T is the sister-in-law of U's mother
- b. P's income is double the difference between R's income and V's income
- c. Both S and R are older than T
- d. Q is the daughter of the one who is the third eldest person
- e. Both the one who is the youngest person and R's sister-in-law have the same gender

Explanations:**1. Questions**

Final arrangement:

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Months	Dates	Persons
January	18	H
	23	I
April	18	G
	23	B
July	18	J
	23	F
September	18	C
	23	E
December	18	D
	23	A

We have,

- J attends the seminar on an even numbered date in the month, having an odd number of days.
- Only two persons attend the seminar between G and the one who attends the seminar immediately after J.
- G and I are adjacent persons but attend the seminar in different months.

From the above conditions, there are three possibilities

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		Case-1	Case-2	Case-3
Months	Dates	Persons	Persons	Persons
January	18	J		
	23		I	
April	18		G	
	23	I		
July	18	G	J	J
	23			
September	18			
	23			I
December	18			G
	23			

Again, we have,

- As many persons attend the seminar before I as after D.
- The one who attends the seminar adjacent to D attends the seminar five months after F.
- Only two months between F and B.

Here, case-2 is split into case-2a.

From the above conditions, case-3 gets eliminated because the one who attends the seminar adjacent to D attends the seminar five months after E, which is not satisfied.

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		Case-1	Case-2	Case-3	Case-2a
Months	Dates	Persons	Persons	Persons	Persons
January	18	J			B/
	23	B/	I		I
April	18	F	G	D	G
	23	I	B		F
July	18	G	J	J	J
	23	B/	F		B/
September	18	D			
	23			I	
December	18		D	G	D
	23				

Again, we have,

- The number of persons attending the seminar between B and A is **one less than** the number of persons attending the seminar between E and H.
- At most one person attends the seminar between C and D.

From the above conditions, case-1 gets eliminated because the number of persons attending the seminar between B and A is **one less than** the number of persons attending the seminar between E and H, which is not satisfied. Also, case-2a gets eliminated because at most one person attends the seminar between C and D, which is not satisfied.

Hence, case-2 shows the final arrangement

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		Case-1	Case-2	Case-2a
Months	Dates	Persons	Persons	Persons
January	18	J	H	C
	23	B	I	I
April	18	F	G	G
	23	I	B	F
July	18	G	J	J
	23		F	B
September	18	D	C	H
	23		E	A
December	18		D	D
	23		A	E

Answer: D

2. Questions

Final arrangement:

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Months	Dates	Persons
January	18	H
	23	I
April	18	G
	23	B
July	18	J
	23	F
September	18	C
	23	E
December	18	D
	23	A

We have,

- J attends the seminar on an even numbered date in the month, having an odd number of days.
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- G and I are adjacent persons but attend the seminar in different months.

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		Case-1	Case-2	Case-3
Months	Dates	Persons	Persons	Persons
January	18	J		
	23		I	
April	18		G	
	23	I		
July	18	G	J	J
	23			
September	18			
	23			I
December	18			G
	23			

Again, we have,

- As many persons attend the seminar before I as after D.
- The one who attends the seminar adjacent to D attends the seminar five months after F.
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Months	Dates	Persons	Persons	Persons	Persons
January	18	J			B/
	23	B/	I		I
April	18	F	G	D	G
	23	I	B		F
July	18	G	J	J	J
	23	B/	F		B/
September	18	D			
	23			I	
December	18		D	G	D
	23				

Again, we have,

- The number of persons attending the seminar between B and A is **one less than** the number of persons attending the seminar between E and H.
- At most one person attends the seminar between C and D.

From the above conditions, case-1 gets eliminated because the number of persons attending the seminar between B and A is **one less than** the number of persons attending the seminar between E and H, which is not satisfied. Also, case-2a gets eliminated because at most one person attends the seminar between C and D, which is not satisfied.

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Months	Dates	Persons	Persons	Persons
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	23	B	I	I
April	18	F	G	G
	23	I	B	F
July	18	G	J	J
	23		F	B
September	18	D	C	H
	23		E	A
December	18		D	D
	23		A	E

Answer: E

3. Questions

Final arrangement:

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Months	Dates	Persons
January	18	H
	23	I
April	18	G
	23	B
July	18	J
	23	F
September	18	C
	23	E
December	18	D
	23	A

We have,

- J attends the seminar on an even numbered date in the month, having an odd number of days.
- Only two persons attend the seminar between G and the one who attends the seminar immediately after J.
- G and I are adjacent persons but attend the seminar in different months.

From the above conditions, there are three possibilities

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		Case-1	Case-2	Case-3
Months	Dates	Persons	Persons	Persons
January	18	J		
	23		I	
April	18		G	
	23	I		
July	18	G	J	J
	23			
September	18			
	23			I
December	18			G
	23			

Again, we have,

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- The one who attends the seminar adjacent to D attends the seminar five months after F.
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Months	Dates	Persons	Persons	Persons	Persons
January	18	J			B/
	23	B/	I		I
April	18	F	G	D	G
	23	I	B		F
July	18	G	J	J	J
	23	B/	F		B/
September	18	D			
	23			I	
December	18		D	G	D
	23				

Again, we have,

- The number of persons attending the seminar between B and A is **one less than** the number of persons attending the seminar between E and H.
- At most one person attends the seminar between C and D.

From the above conditions, case-1 gets eliminated because the number of persons attending the seminar between B and A is **one less than** the number of persons attending the seminar between E and H, which is not satisfied. Also, case-2a gets eliminated because at most one person attends the seminar between C and D, which is not satisfied.

Hence, case-2 shows the final arrangement

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		Case-1	Case-2	Case-2a
Months	Dates	Persons	Persons	Persons
January	18	J	H	C
	23	B	I	I
April	18	F	G	G
	23	I	B	F
July	18	G	J	J
	23		F	B
September	18	D	C	H
	23		E	A
December	18		D	D
	23		A	E

Answer: C

4. Questions

Final arrangement:

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Months	Dates	Persons
January	18	H
	23	I
April	18	G
	23	B
July	18	J
	23	F
September	18	C
	23	E
December	18	D
	23	A

We have,

- J attends the seminar on an even numbered date in the month, having an odd number of days.
- Only two persons attend the seminar between G and the one who attends the seminar immediately after J.
- G and I are adjacent persons but attend the seminar in different months.

From the above conditions, there are three possibilities

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		Case-1	Case-2	Case-3
Months	Dates	Persons	Persons	Persons
January	18	J		
	23		I	
April	18		G	
	23	I		
July	18	G	J	J
	23			
September	18			
	23			I
December	18			G
	23			

Again, we have,

- As many persons attend the seminar before I as after D.
- The one who attends the seminar adjacent to D attends the seminar five months after F.
- Only two months between F and B.

Here, case-2 is split into case-2a.

From the above conditions, case-3 gets eliminated because the one who attends the seminar adjacent to D attends the seminar five months after E, which is not satisfied.

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Months	Dates	Persons	Persons	Persons	Persons
January	18	J			B/
	23	B/	I		I
April	18	F	G	D	G
	23	I	B		F
July	18	G	J	J	J
	23	B/	F		B/
September	18	D			
	23			I	
December	18		D	G	D
	23				

Again, we have,

- The number of persons attending the seminar between B and A is **one less than** the number of persons attending the seminar between E and H.
- At most one person attends the seminar between C and D.

From the above conditions, case-1 gets eliminated because the number of persons attending the seminar between B and A is **one less than** the number of persons attending the seminar between E and H, which is not satisfied. Also, case-2a gets eliminated because at most one person attends the seminar between C and D, which is not satisfied.

Hence, case-2 shows the final arrangement

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		Case-1	Case-2	Case-2a
Months	Dates	Persons	Persons	Persons
January	18	J	H	C
	23	B	I	I
April	18	F	G	G
	23	I	B	F
July	18	G	J	J
	23		F	B
September	18	D	C	H
	23		E	A
December	18		D	D
	23		A	E

Answer: A

5. Questions

Final arrangement:

@QuizByMickey

Months	Dates	Persons
January	18	H
	23	I
April	18	G
	23	B
July	18	J
	23	F
September	18	C
	23	E
December	18	D
	23	A

We have,

- J attends the seminar on an even numbered date in the month, having an odd number of days.
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		Case-1	Case-2	Case-3
Months	Dates	Persons	Persons	Persons
January	18	J		
	23		I	
April	18		G	
	23	I		
July	18	G	J	J
	23			
September	18			
	23			I
December	18			G
	23			

Again, we have,

- As many persons attend the seminar before I as after D.
- The one who attends the seminar adjacent to D attends the seminar five months after F.
- Only two months between F and B.

Here, case-2 is split into case-2a.

From the above conditions, case-3 gets eliminated because the one who attends the seminar adjacent to D attends the seminar five months after E, which is not satisfied.

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		Case-1	Case-2	Case-3	Case-2a
Months	Dates	Persons	Persons	Persons	Persons
January	18	J			B/
	23	B/	I		I
April	18	F	G	D	G
	23	I	B		F
July	18	G	J	J	J
	23	B/	F		B/
September	18	D			
	23			I	
December	18		D	G	D
	23				

Again, we have,

- The number of persons attending the seminar between B and A is **one less than** the number of persons attending the seminar between E and H.
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From the above conditions, case-1 gets eliminated because the number of persons attending the seminar between B and A is **one less than** the number of persons attending the seminar between E and H, which is not satisfied. Also, case-2a gets eliminated because at most one person attends the seminar between C and D, which is not satisfied.

Hence, case-2 shows the final arrangement

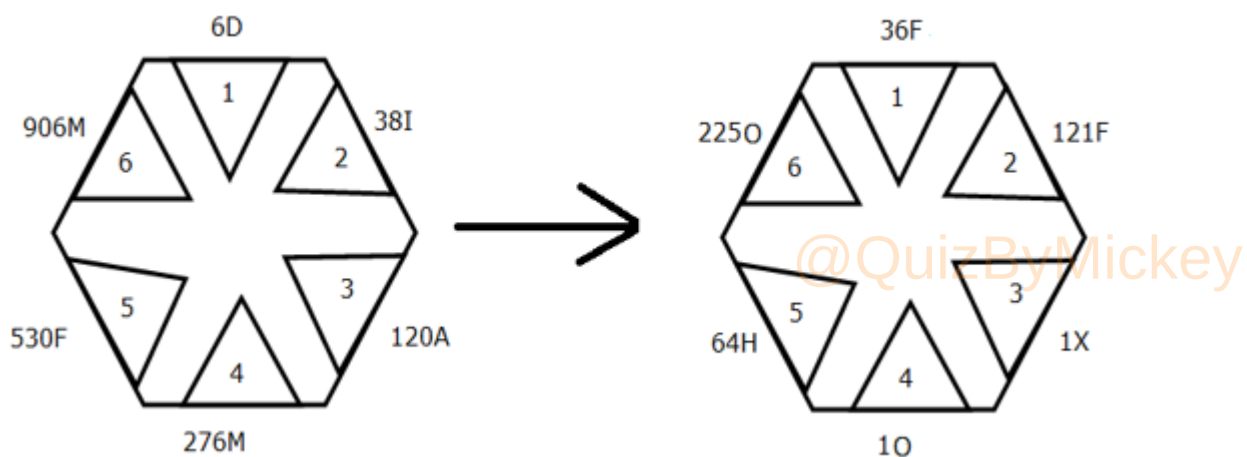
@QuizByMickey

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Months	Dates	Persons	Persons	Persons
January	18	J	H	C
	23	B	I	I
April	18	F	G	G
	23	I	B	F
July	18	G	J	J
	23		F	B
September	18	D	C	H
	23		E	A
December	18		D	D
	23		A	E

Answer: B

6. Questions

Final arrangement:



Input of the triangle 1 to 6:

The value for each triangle is the difference between the square value of the first six odd numbers (except 1) and the first six numbers divisible by three. Then, the resultant value is multiplied with the first six natural numbers and then the values are arranged in order from triangles 1 to 6, respectively.

$$(3)^2=9-3=6 \times 1=6$$

$$(5)^2=25-6=19 \times 2=38$$

$$(7)^2=49-9=40 \times 3=120$$

$$(9)^2=81-12=69 \times 4=276$$

$$(11)^2=121-15=106 \times 5=530$$

$$(13)^2=169-18=151 \times 6=906$$

The values are 6, 38, 120, 276, 530, 906.

The letter for each triangle is the corresponding letter of the place value, which is the sum of the digits of the value of each triangle, thus obtained as per the English alphabetical series. Then each obtained letter is changed into its second preceding letter. Then the letters are arranged from triangles 1 to 6, respectively.

$$6-F \rightarrow D$$

$$3+8=11-K \rightarrow I$$

$$1+2+0=3-C \rightarrow A$$

$$2+7+6=15-O \rightarrow M$$

$$5+3+0=8-H \rightarrow F$$

$$9+0+6=15-O \rightarrow M$$

The values and the letters are arranged from triangle 1 to 6

6D, 38I, 120A, 276M, 530F, 906M

Output:

I). If the triangle has a consonant letter, then the letter is replaced by the second succeeding letter as per the English alphabetical series.

II). If the triangle has a vowel letter, then the letter is replaced by the third preceding letter as per the English alphabetical series.

D, I, A, M, F, M

F, F, X, O, H, O

III). If the triangle has values divisible by 4, then the difference between the sum of the odd digits and the sum of the even digits is taken.

Only two values, 120 and 276, are divisible by 4

$$(120)$$

$$2-1=1$$

$$(276)$$

$$2+6-7=1$$

$$120 \rightarrow 1$$

276 → 1

IV). If the triangle has a value not divisible by 4, then the value is the square of the sum of the digits of the number.

6, 38, 530, 906, these are not divisible by 4.

$(6)^2=36$

$3+8=(11)^2=121$

$5+3+0=(8)^2=64$

$9+0+6=(15)^2=225$

6 → 36

38 → 121

530 → 64

906 → 225

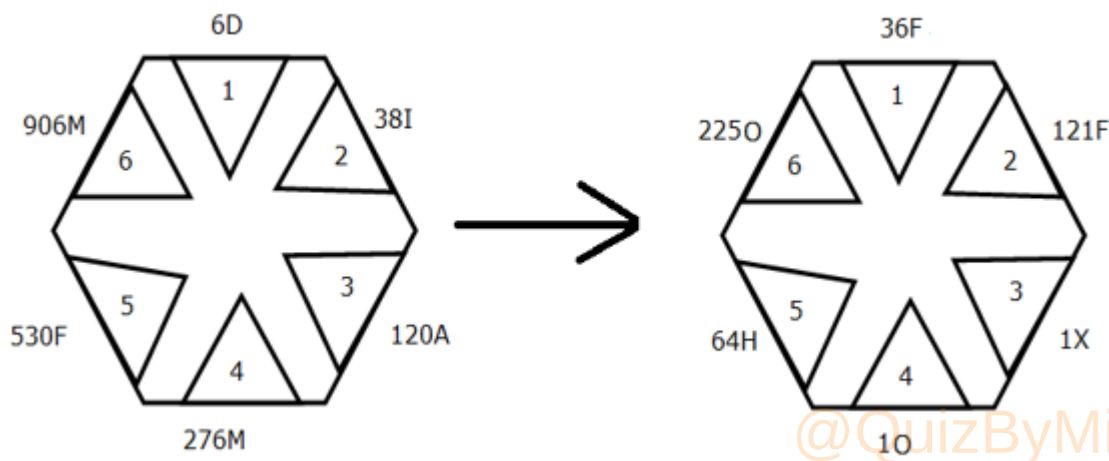
The final output of the triangle 1 to 6

36F, 121F, 1X, 1O, 64H, 225P

Answer: E

7. Questions

Final arrangement:



Input of the triangle 1 to 6:

The value for each triangle is the difference between the square value of the first six odd numbers (except 1) and the first six numbers divisible by three. Then, the resultant value is multiplied with the first six natural numbers and then the values are arranged in order from triangles 1 to 6, respectively.

$(3)^2=9-3=6 \times 1=6$

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The values are 6, 38, 120, 276, 530, 906.

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The values and the letters are arranged from triangle 1 to 6

6D, 38I, 120A, 276M, 530F, 906M

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F, F, X, O, H, O

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Only two values, 120 and 276, are divisible by 4

(120)

$$2-1=1$$

(276)

$$2+6-7=1$$

120 $\rightarrow$ 1

276 $\rightarrow$ 1

IV). If the triangle has a value not divisible by 4, then the value is the square of the sum of the digits of the number.

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6, 38, 530, 906, these are not divisible by 4.

$$(6)^2 - 36$$

$$3 + 8 = (11)^2 = 121$$

$$5 + 3 + 0 = (8)^2 = 64$$

$$9 + 0 + 6 = (15)^2 = 225$$

$$6 \rightarrow 36$$

$$38 \rightarrow 121$$

$$530 \rightarrow 64$$

$$906 \rightarrow 225$$

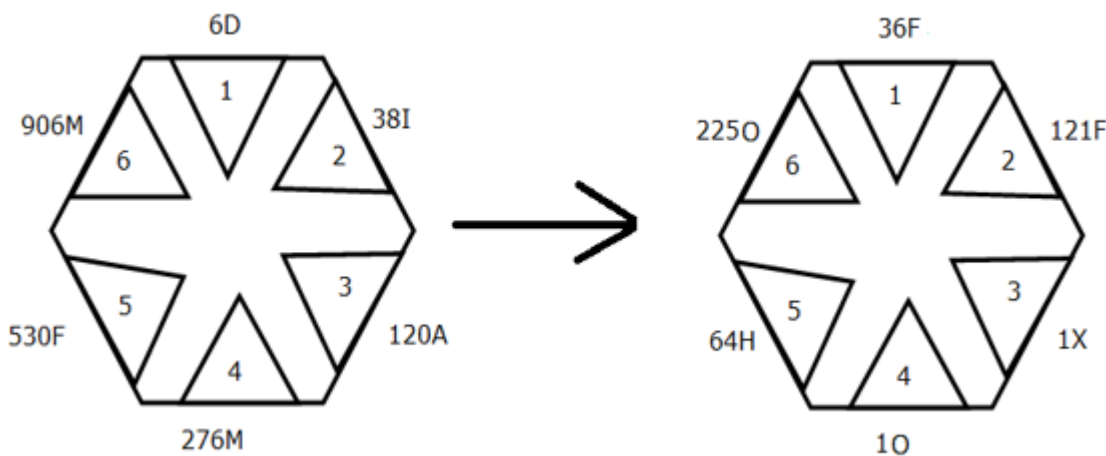
The final output of the triangle 1 to 6

36F, 121F, 1X, 10, 64H, 225P

Answer: B

8. Questions

Final arrangement:



Input of the triangle 1 to 6:

The value for each triangle is the difference between the square value of the first six odd numbers (except 1) and the first six numbers divisible by three. Then, the resultant value is multiplied with the first six natural numbers and then the values are arranged in order from triangles 1 to 6, respectively.

$$(3)^2 - 9 - 3 = 6 \times 1 = 6$$

$$(5)^2 - 25 - 6 = 19 \times 2 = 38$$

$$(7)^2 - 49 - 9 = 40 \times 3 = 120$$

$$(9)^2 - 81 - 12 = 69 \times 4 = 276$$

$$(11)^2 - 121 - 15 = 106 \times 5 = 530$$

$$(13)^2 - 169 - 18 = 151 \times 6 = 906$$

The values are 6, 38, 120, 276, 530, 906.

The letter for each triangle is the corresponding letter of the place value, which is the sum of the digits of the value of each triangle, thus obtained as per the English alphabetical series. Then each obtained letter is changed into its second preceding letter. Then the letters are arranged from triangles 1 to 6, respectively.

6-F→D

3+8=11-K→I

1+2+0=3-C→A

2+7+6=15-O→M

5+3+0=8-H→F

9+0+6=15-O→M

The values and the letters are arranged from triangle 1 to 6

6D, 38I, 120A, 276M, 530F, 906M

Output:

I). If the triangle has a consonant letter, then the letter is replaced by the second succeeding letter as per the English alphabetical series.

II). If the triangle has a vowel letter, then the letter is replaced by the third preceding letter as per the English alphabetical series.

D, I, A, M, F, M

F, F, X, O, H, O

III). If the triangle has values divisible by 4, then the difference between the sum of the odd digits and the sum of the even digits is taken.

Only two values, 120 and 276, are divisible by 4

(120)

2-1=1

(276)

2+6-7=1

120 → 1

276 → 1

IV). If the triangle has a value not divisible by 4, then the value is the square of the sum of the digits of the number.

6, 38, 530, 906, these are not divisible by 4.

(6)²=36

3+8=(11)²=121

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$$5+3+0=(8)^2=64$$

$$9+0+6=(15)^2=225$$

$$6 \rightarrow 36$$

$$38 \rightarrow 121$$

$$530 \rightarrow 64$$

$$906 \rightarrow 225$$

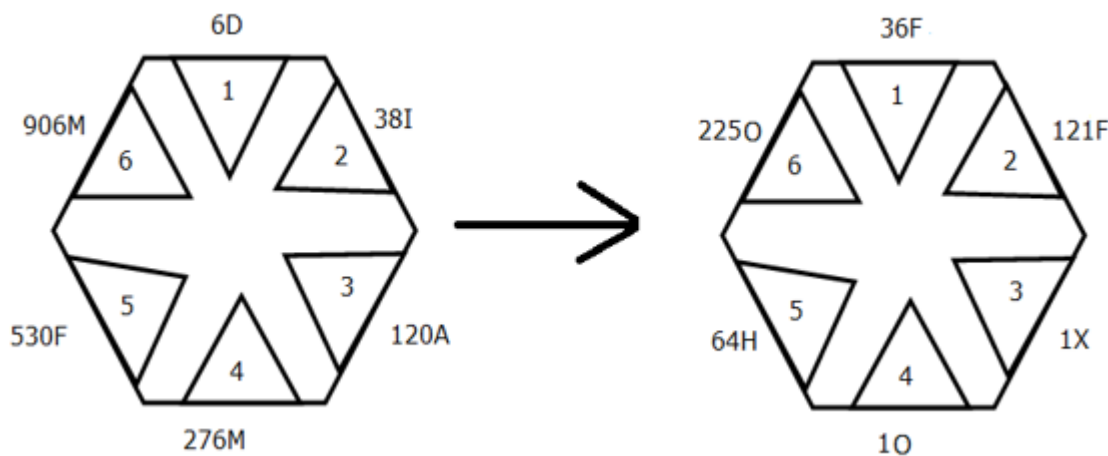
The final output of the triangle 1 to 6

36F, 121F, 1X, 10, 64H, 225P

Answer: D

9. Questions

Final arrangement:



Input of the triangle 1 to 6:

The value for each triangle is the difference between the square value of the first six odd numbers (except 1) and the first six numbers divisible by three. Then, the resultant value is multiplied with the first six natural numbers and then the values are arranged in order from triangles 1 to 6, respectively.

$$(3)^2=9-3=6 \times 1=6$$

$$(5)^2=25-6=19 \times 2=38$$

$$(7)^2=49-9=40 \times 3=120$$

$$(9)^2=81-12=69 \times 4=276$$

$$(11)^2=121-15=106 \times 5=530$$

$$(13)^2=169-18=151 \times 6=906$$

The values are 6, 38, 120, 276, 530, 906.

The letter for each triangle is the corresponding letter of the place value, which is the sum of the digits of the value of each triangle, thus obtained as per the English alphabetical series. Then each obtained letter is

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changed into its second preceding letter. Then the letters are arranged from triangles 1 to 6, respectively.

6-F→D

3+8=11-K→I

1+2+0=3-C→A

2+7+6=15-O→M

5+3+0=8-H→F

9+0+6=15-O→M

The values and the letters are arranged from triangle 1 to 6

6D, 38I, 120A, 276M, 530F, 906M

Output:

I). If the triangle has a consonant letter, then the letter is replaced by the second succeeding letter as per the English alphabetical series.

II). If the triangle has a vowel letter, then the letter is replaced by the third preceding letter as per the English alphabetical series.

D, I, A, M, F, M

F, F, X, O, H, O

III). If the triangle has values divisible by 4, then the difference between the sum of the odd digits and the sum of the even digits is taken.

Only two values, 120 and 276, are divisible by 4

(120)

2-1=1

(276)

2+6-7=1

120 →> 1

276 →>1

IV). If the triangle has a value not divisible by 4, then the value is the square of the sum of the digits of the number.

6, 38, 530, 906, these are not divisible by 4.

(6)^2=36

3+8=(11)^2=121

5+3+0=(8)^2=64

9+0+6=(15)^2=225

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6—>36

38—>121

530—>64

906—>225

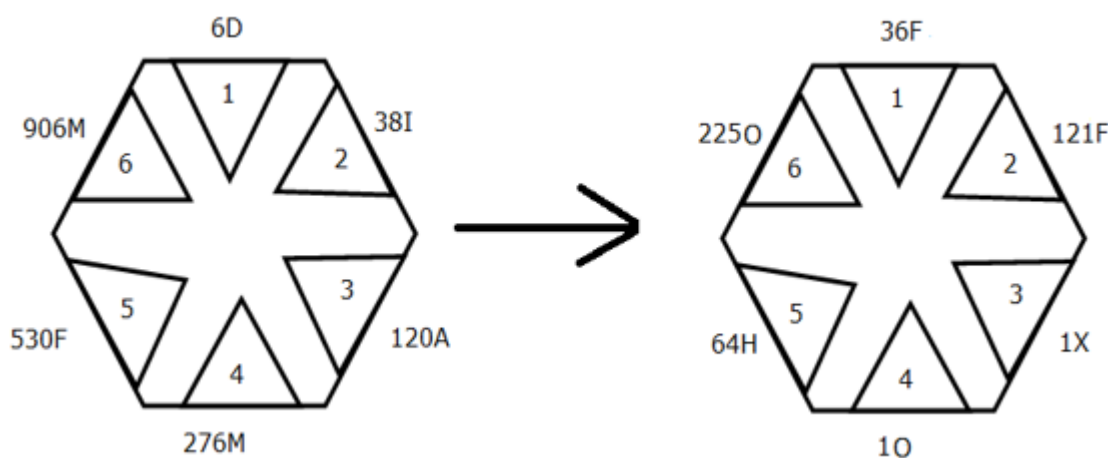
The final output of the triangle 1 to 6

36F, 121F, 1X, 1O, 64H, 225P

Answer: D

10. Questions

Final arrangement:



Input of the triangle 1 to 6:

The value for each triangle is the difference between the square value of the first six odd numbers (except 1) and the first six numbers divisible by three. Then, the resultant value is multiplied with the first six natural numbers and then the values are arranged in order from triangles 1 to 6, respectively.

$$(3)^2=9-3=6 \times 1=6$$

$$(5)^2=25-6=19 \times 2=38$$

$$(7)^2=49-9=40 \times 3=120$$

$$(9)^2=81-12=69 \times 4=276$$

$$(11)^2=121-15=106 \times 5=530$$

$$(13)^2=169-18=151 \times 6=906$$

The values are 6, 38, 120, 276, 530, 906.

The letter for each triangle is the corresponding letter of the place value, which is the sum of the digits of the value of each triangle, thus obtained as per the English alphabetical series. Then each obtained letter is changed into its second preceding letter. Then the letters are arranged from triangles 1 to 6, respectively.

6-F→D

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$$3+8=11-K \rightarrow I$$

$$1+2+0=3-C \rightarrow A$$

$$2+7+6=15-O \rightarrow M$$

$$5+3+0=8-H \rightarrow F$$

$$9+0+6=15-O \rightarrow M$$

The values and the letters are arranged from triangle 1 to 6

6D, 38I, 120A, 276M, 530F, 906M

Output:

I). If the triangle has a consonant letter, then the letter is replaced by the second succeeding letter as per the English alphabetical series.

II). If the triangle has a vowel letter, then the letter is replaced by the third preceding letter as per the English alphabetical series.

D, I, A, M, F, M

F, F, X, O, H, O

III). If the triangle has values divisible by 4, then the difference between the sum of the odd digits and the sum of the even digits is taken.

Only two values, 120 and 276, are divisible by 4

(120)

$$2-1=1$$

(276)

$$2+6-7=1$$

$$120 \rightarrow 1$$

$$276 \rightarrow 1$$

IV). If the triangle has a value not divisible by 4, then the value is the square of the sum of the digits of the number.

6, 38, 530, 906, these are not divisible by 4.

$$(6)^2=36$$

$$3+8=(11)^2=121$$

$$5+3+0=(8)^2=64$$

$$9+0+6=(15)^2=225$$

$$6 \rightarrow 36$$

$$38 \rightarrow 121$$

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530—>64

906—>225

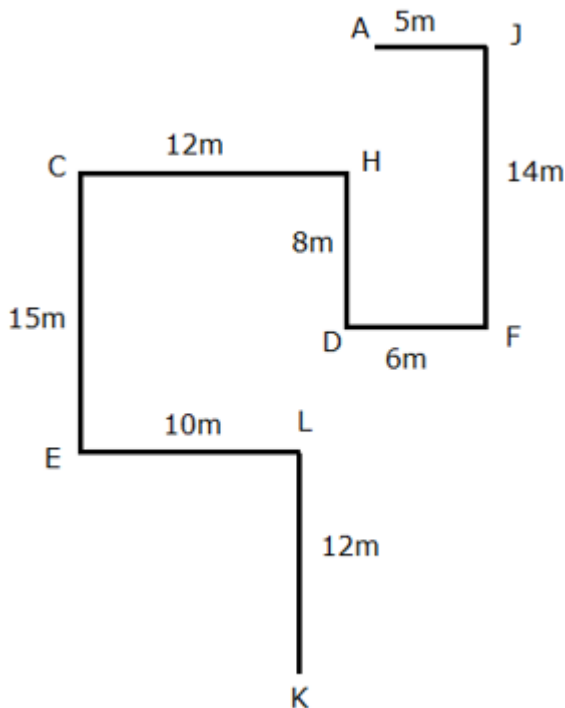
The final output of the triangle 1 to 6

36F, 121F, 1X, 1O, 64H, 225P

Answer: C

11. Questions

Final arrangement:



“A#B(12)” means “A is 15m to the north of B”

If the symbol has A#B(12) means add 3m to that of value

“A@B(17)” means “A is 14m to the south of B”.

If the symbol has “A@B(17)” means subtract 3m to that of value

“A&B(21)” means “A is 23m west of B”.

If the symbol has “A&B(21)” means add 2m to that of value

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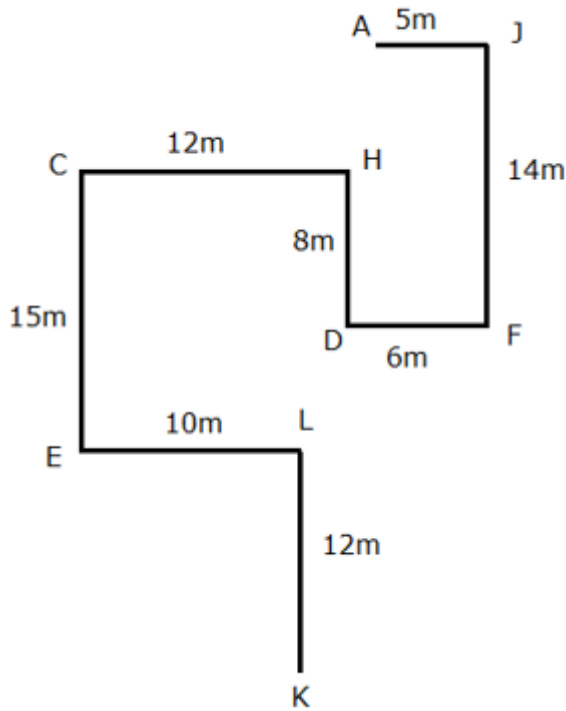
“A^B(24)” means “A is 20m east of B”.

If the symbol has “A^B(24)” means subtract 4m to that of value

Answer: C

12. Questions

Final arrangement:



“A#B(12)” means “A is 15m to the north of B”

If the symbol has A#B(12) means add 3m to that of value

“A@B(17)” means “A is 14m to the south of B”.

If the symbol has “A@B(17)” means subtract 3m to that of value

“A&B(21)” means “A is 23m west of B”.

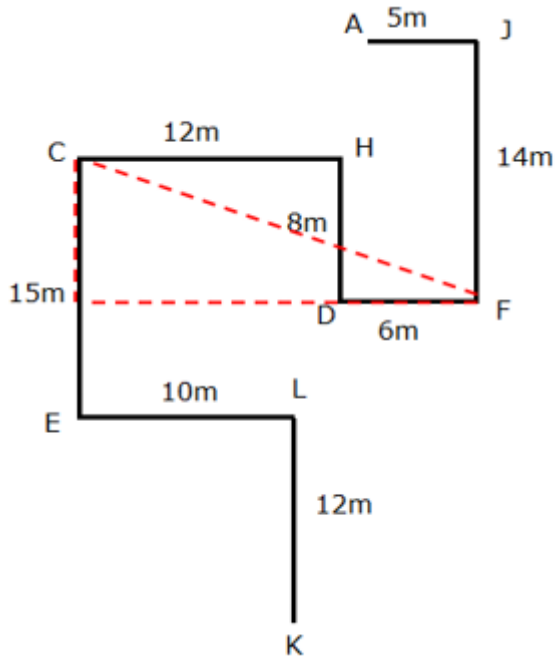
If the symbol has “A&B(21)” means add 2m to that of value

“A^B(24)” means “A is 20m east of B”.

If the symbol has “A^B(24)” means subtract 4m to that of value

Answer: D

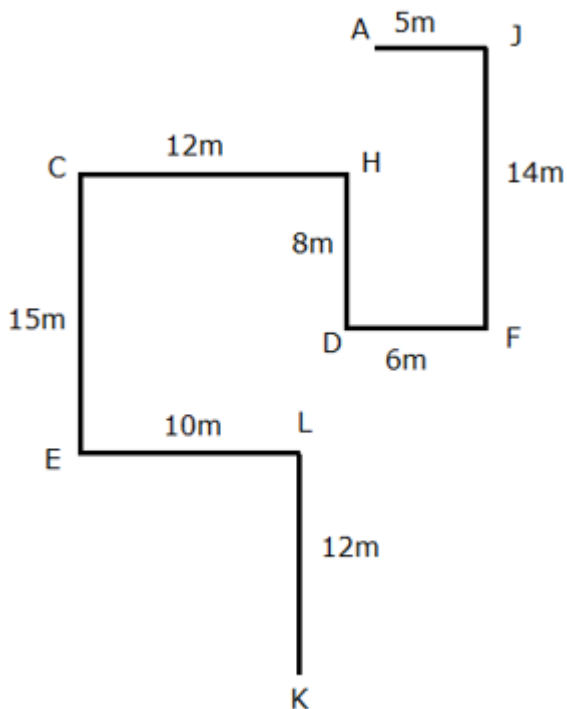
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$$\sqrt{18^2 + 8^2} = \sqrt{324 + 64} = \sqrt{388} \text{m}$$

13. Questions

Final arrangement:



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“A#B(12)” means “A is 15m to the north of B”

If the symbol has A#B(12) means add 3m to that of value

“A@B(17)” means “A is 14m to the south of B”.

If the symbol has “A@B(17)” means subtract 3m to that of value

“A&B(21)” means “A is 23m west of B”.

If the symbol has “A&B(21)” means add 2m to that of value

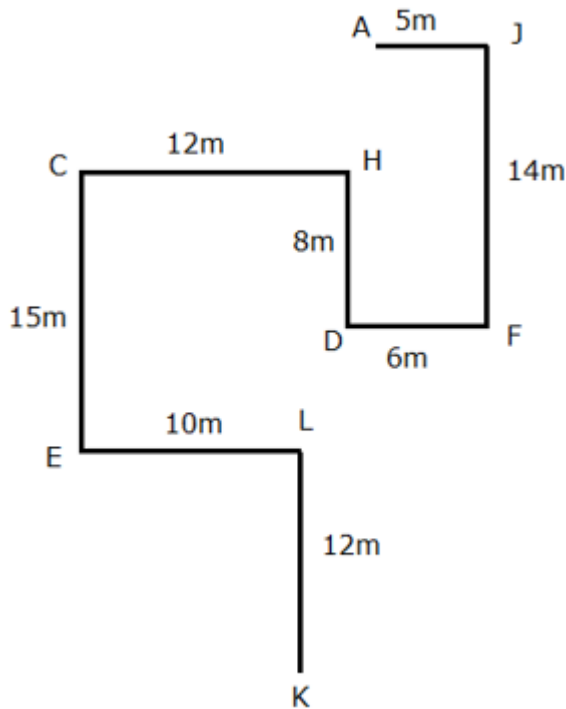
“A^B(24)” means “A is 20m east of B”.

If the symbol has “A^B(24)” means subtract 4m to that of value

Answer: B (In all other options, the first point is the north-east to the second point except option b)

14. Questions

Final arrangement:



“A#B(12)” means “A is 15m to the north of B”

If the symbol has A#B(12) means add 3m to that of value

“A@B(17)” means “A is 14m to the south of B”.

If the symbol has “A@B(17)” means subtract 3m to that of value

“A&B(21)” means “A is 23m west of B”.

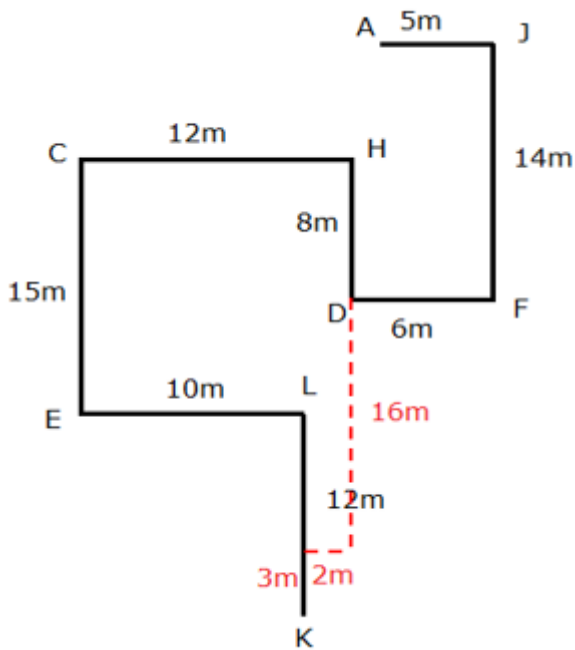
If the symbol has “A&B(21)” means add 2m to that of value

“A^B(24)” means “A is 20m east of B”.

If the symbol has “A^B(24)” means subtract 4m to that of value

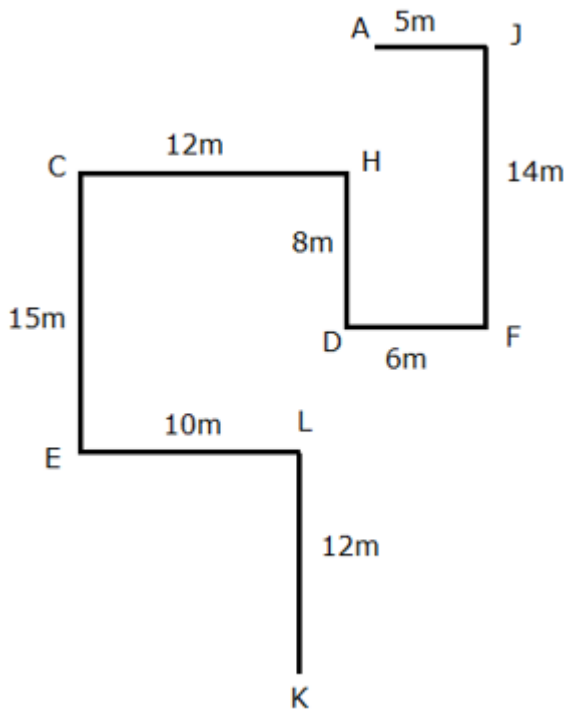
Answer: A

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15. Questions

Final arrangement:



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“A#B(12)” means “A is 15m to the north of B”

If the symbol has A#B(12) means add 3m to that of value

“A@B(17)” means “A is 14m to the south of B”.

If the symbol has “A@B(17)” means subtract 3m to that of value

“A&B(21)” means “A is 23m west of B”.

If the symbol has “A&B(21)” means add 2m to that of value

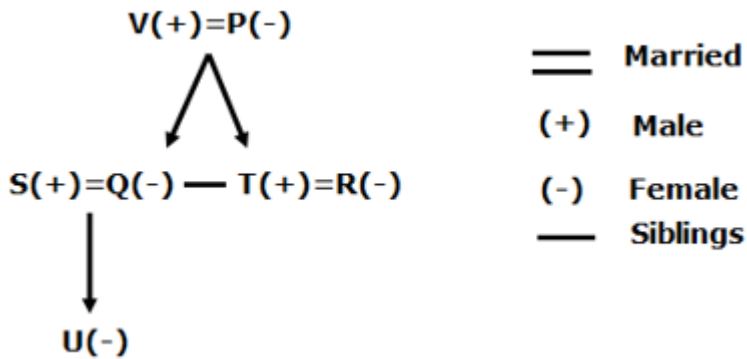
“A^B(24)” means “A is 20m east of B”.

If the symbol has “A^B(24)” means subtract 4m to that of value

Answer: E

16. Questions

Final arrangement:



$V(48K) > P(52k) > S(41k) > T(47\text{years}) > Q(42\text{years})(37k) > R(29k) > U(15\text{years})$

We have,

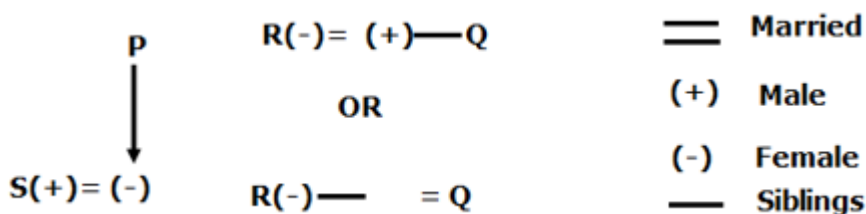
- R is the sister-in-law of Q and is younger than T.
- Q earns 37k and is 27 years older than U.
- S is the son-in-law of P and older than R.

Explanation,

$T > R, S > R$

$Q = 27 + U$

$Q(37k) > U$



Again, we have,

- V earns 7k more than S.
- T is the brother-in-law of S, who doesn't have any siblings. T is 5 years older than Q.
- V is the father-in-law of R and earns 11k more than Q.

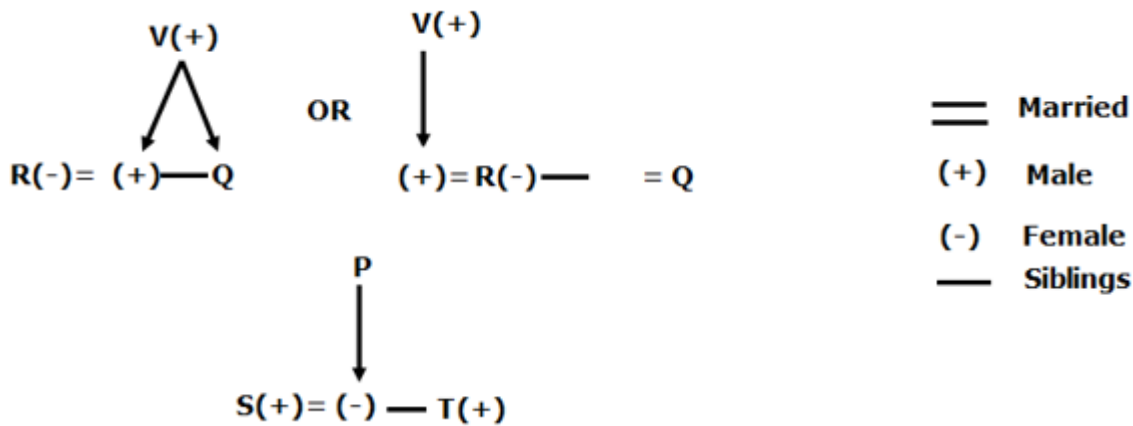
Explanation,

$V = 7k + S \rightarrow S = 41k$

$T = 5 + Q (T > Q)$

$V = 11k + Q \rightarrow V = 48k$

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Again, we have,

- T is the maternal uncle of U and just younger than S.
- P earns 4k more than V and is older than S.
- R earns 23k less than P and is younger than Q and is older than U, who is the granddaughter of V.
- U's age is 15 years.

Explanation,

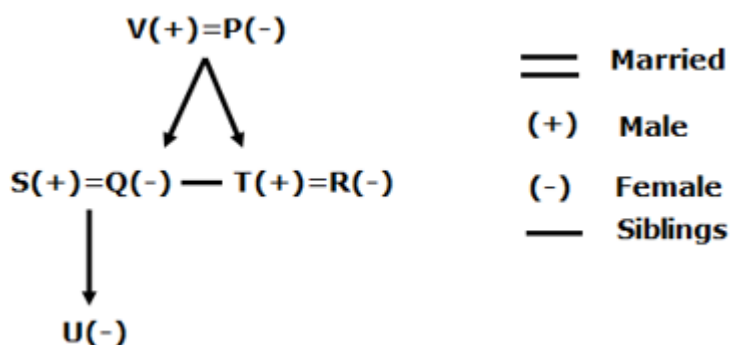
$$P > S > T > R$$

$$P = 48k + 4k = 52k$$

$$R = 52k - 23k = 29k$$

$$Q > R > U \text{ (15year)}$$

$$V(48K) > P(52k) > S(41k) > T(47years) > Q(42years)(37k) > R(29k) > U \text{ (15years)}$$

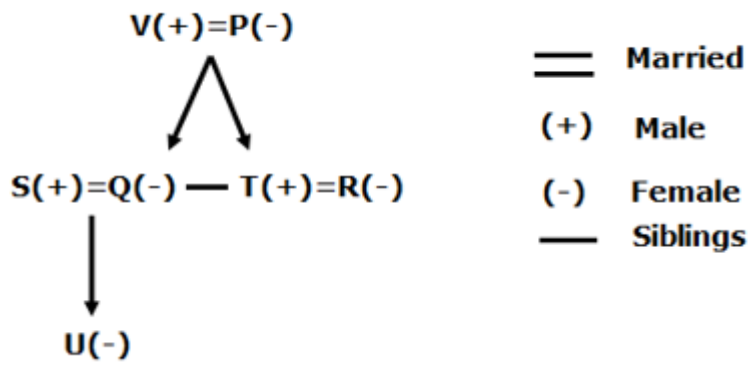


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Answer: C

17. Questions

Final arrangement:



$V(48K) > P(52k) > S(41k) > T(47\text{years}) > Q(42\text{years})(37k) > R(29k) > U(15\text{years})$

We have,

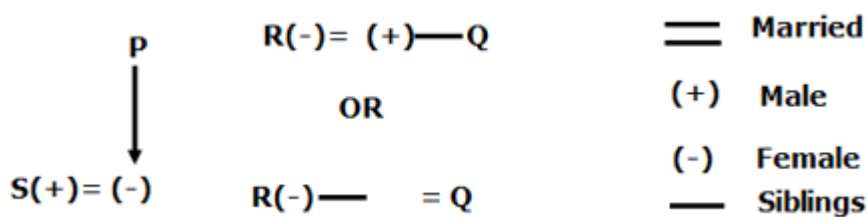
- R is the sister-in-law of Q and is younger than T.
- Q earns 37k and is 27 years older than U.
- S is the son-in-law of P and older than R.

Explanation,

$T > R, S > R$

$Q = 27 + U$

$Q(37k) > U$



Again, we have,

- V earns 7k more than S.
- T is the brother-in-law of S, who doesn't have any siblings. T is 5 years older than Q.
- V is the father-in-law of R and earns 11k more than Q.

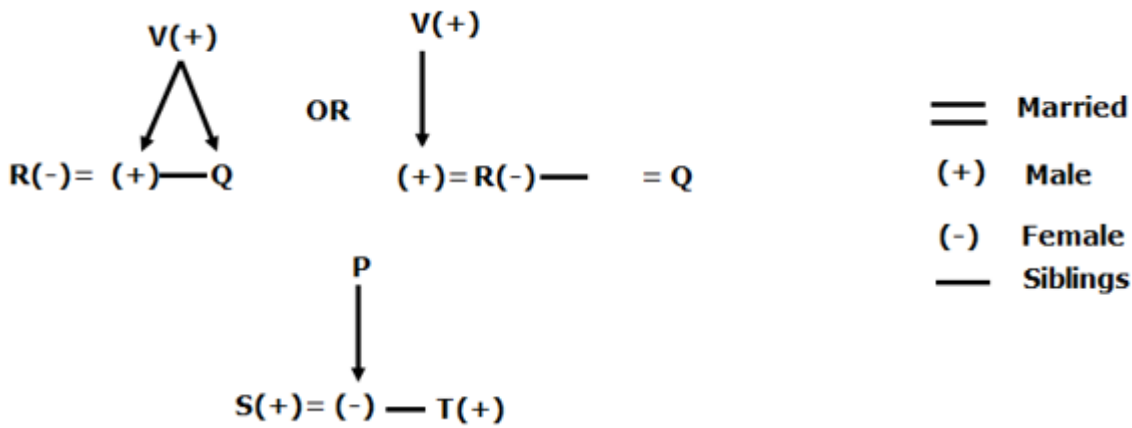
Explanation,

$V = 7k + S \rightarrow S = 41k$

$T = 5 + Q (T > Q)$

$V = 11k + Q \rightarrow V = 48k$

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Again, we have,

- T is the maternal uncle of U and just younger than S.
- P earns 4k more than V and is older than S.
- R earns 23k less than P and is younger than Q and is older than U, who is the granddaughter of V.
- U's age is 15 years.

Explanation,

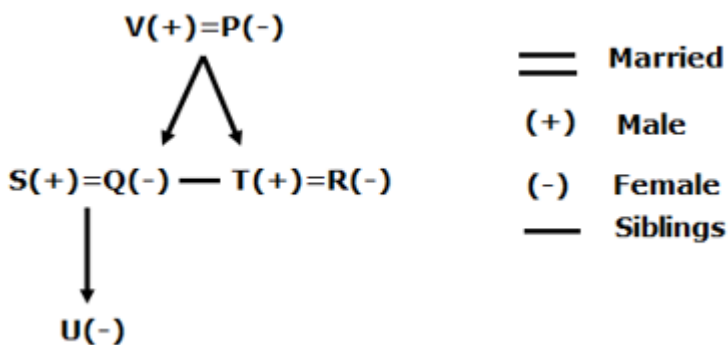
$$P > S > T > R$$

$$P = 48k + 4k = 52k$$

$$R = 52k - 23k = 29k$$

$$Q > R > U \text{ (15year)}$$

$$V(48K) > P(52k) > S(41k) > T(47years) > Q(42years)(37k) > R(29k) > U \text{ (15years)}$$

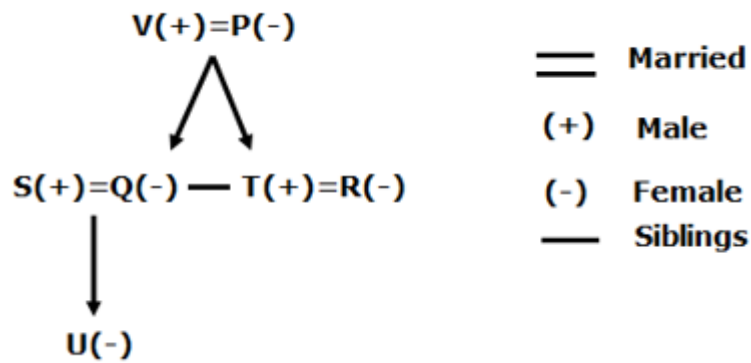


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Answer: B

18. Questions

Final arrangement:



$V(48K) > P(52k) > S(41k) > T(47\text{years}) > Q(42\text{years})(37k) > R(29k) > U(15\text{years})$

We have,

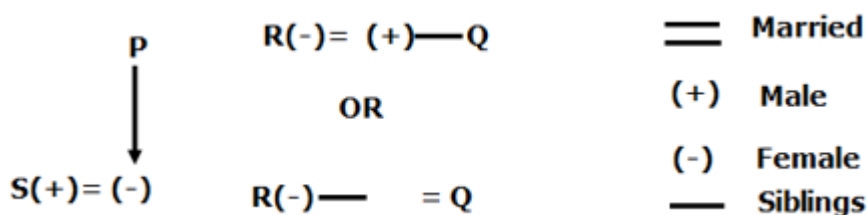
- R is the sister-in-law of Q and is younger than T.
- Q earns 37k and is 27 years older than U.
- S is the son-in-law of P and older than R.

Explanation,

$T > R, S > R$

$Q = 27 + U$

$Q(37k) > U$



Again, we have,

- V earns 7k more than S.
- T is the brother-in-law of S, who doesn't have any siblings. T is 5 years older than Q.
- V is the father-in-law of R and earns 11k more than Q.

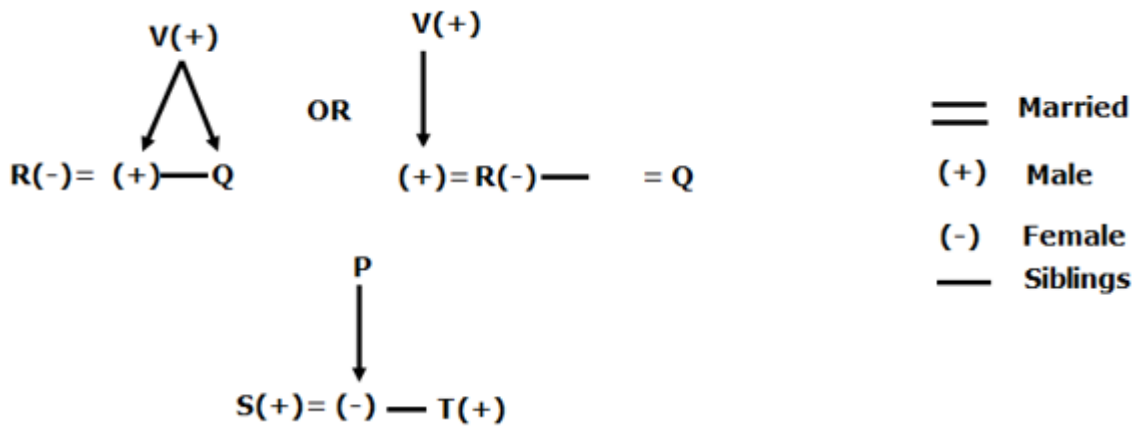
Explanation,

$V = 7k + S \rightarrow S = 41k$

$T = 5 + Q (T > Q)$

$V = 11k + Q \rightarrow V = 48k$

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Again, we have,

- T is the maternal uncle of U and just younger than S.
- P earns 4k more than V and is older than S.
- R earns 23k less than P and is younger than Q and is older than U, who is the granddaughter of V.
- U's age is 15 years.

Explanation,

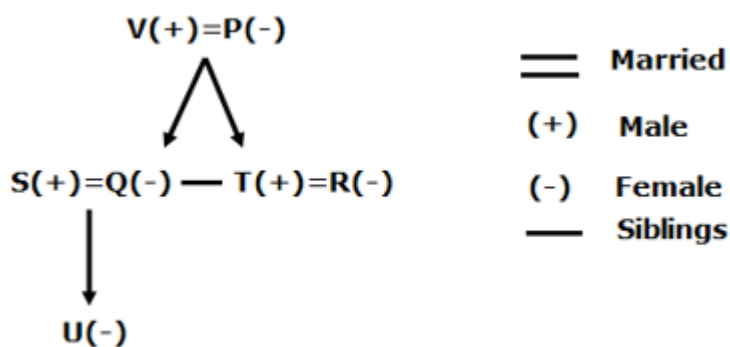
$$P > S > T > R$$

$$P = 48k + 4k = 52k$$

$$R = 52k - 23k = 29k$$

$$Q > R > U \text{ (15year)}$$

$$V(48K) > P(52k) > S(41k) > T(47years) > Q(42years)(37k) > R(29k) > U \text{ (15years)}$$

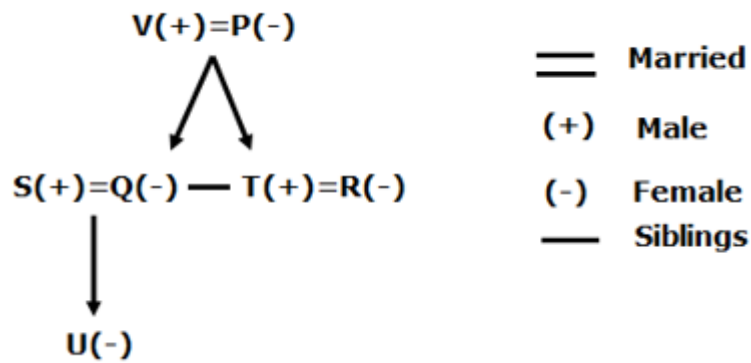


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Answer: D

19. Questions

Final arrangement:



$V(48K) > P(52k) > S(41k) > T(47\text{years}) > Q(42\text{years})(37k) > R(29k) > U(15\text{years})$

We have,

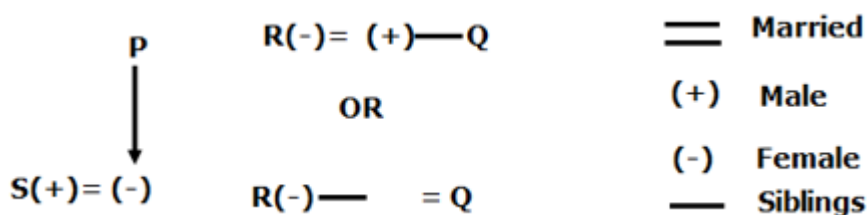
- R is the sister-in-law of Q and is younger than T.
- Q earns 37k and is 27 years older than U.
- S is the son-in-law of P and older than R.

Explanation,

$T > R, S > R$

$Q = 27 + U$

$Q(37k) > U$



Again, we have,

- V earns 7k more than S.
- T is the brother-in-law of S, who doesn't have any siblings. T is 5 years older than Q.
- V is the father-in-law of R and earns 11k more than Q.

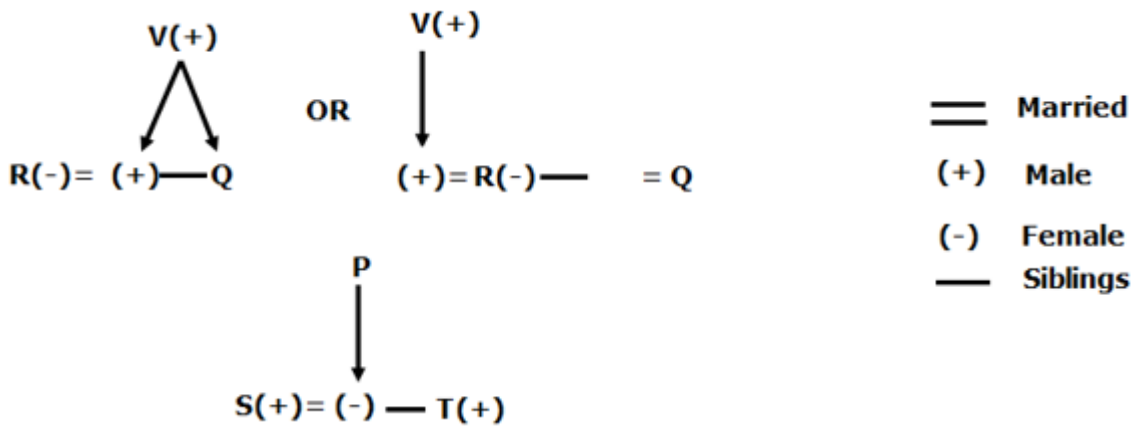
Explanation,

$V = 7k + S \rightarrow S = 41k$

$T = 5 + Q (T > Q)$

$V = 11k + Q \rightarrow V = 48k$

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Again, we have,

- T is the maternal uncle of U and just younger than S.
- P earns 4k more than V and is older than S.
- R earns 23k less than P and is younger than Q and is older than U, who is the granddaughter of V.
- U's age is 15 years.

Explanation,

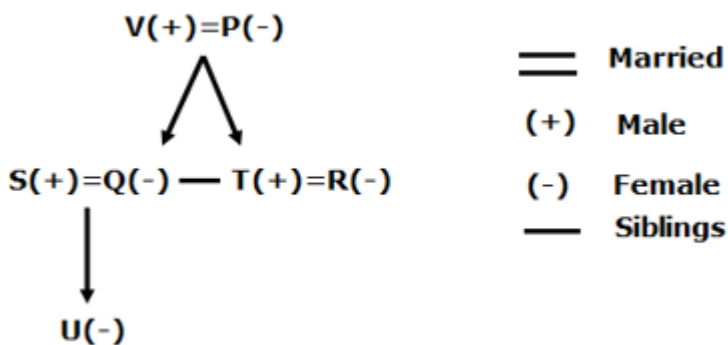
$$P > S > T > R$$

$$P = 48k + 4k = 52k$$

$$R = 52k - 23k = 29k$$

$$Q > R > U \text{ (15 year)}$$

$$V(48K) > P(52k) > S(41k) > T(47\text{years}) > Q(42\text{years})(37k) > R(29k) > U \text{ (15 years)}$$

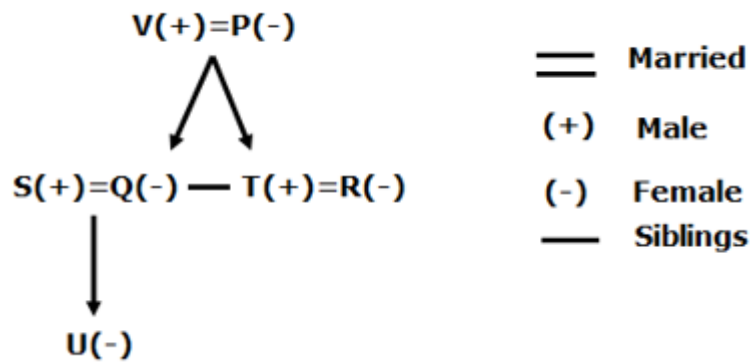


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Answer: A

20. Questions

Final arrangement:



$V(48K) > P(52k) > S(41k) > T(47\text{years}) > Q(42\text{years})(37k) > R(29k) > U(15\text{years})$

We have,

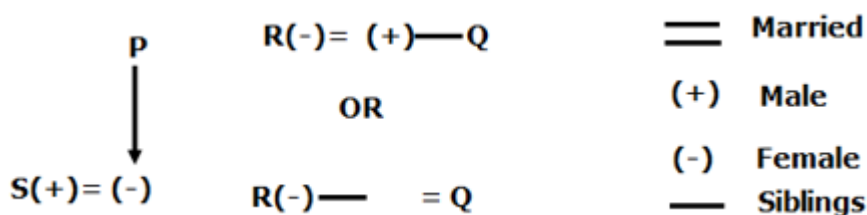
- R is the sister-in-law of Q and is younger than T.
- Q earns 37k and is 27 years older than U.
- S is the son-in-law of P and older than R.

Explanation,

$T > R, S > R$

$Q = 27 + U$

$Q(37k) > U$



Again, we have,

- V earns 7k more than S.
- T is the brother-in-law of S, who doesn't have any siblings. T is 5 years older than Q.
- V is the father-in-law of R and earns 11k more than Q.

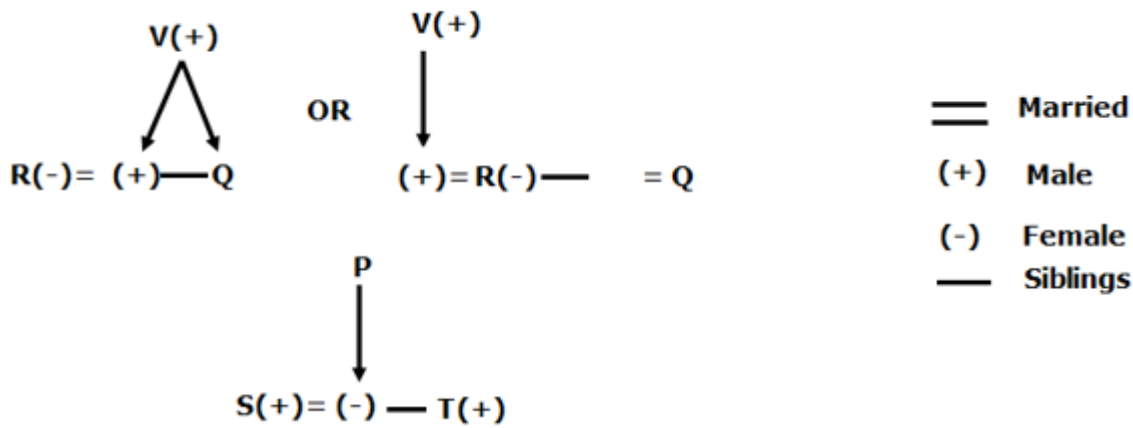
Explanation,

$V = 7k + S \rightarrow S = 41k$

$T = 5 + Q (T > Q)$

$V = 11k + Q \rightarrow V = 48k$

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Again, we have,

- T is the maternal uncle of U and just younger than S.
- P earns 4k more than V and is older than S.
- R earns 23k less than P and is younger than Q and is older than U, who is the granddaughter of V.
- U's age is 15 years.

Explanation,

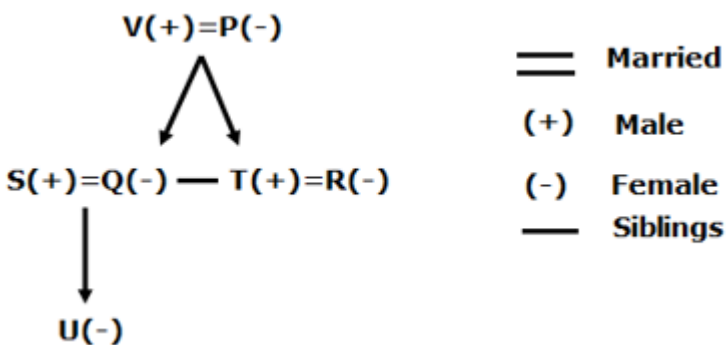
$$P > S > T > R$$

$$P = 48k + 4k = 52k$$

$$R = 52k - 23k = 29k$$

$$Q > R > U \text{ (15year)}$$

$$V(48K) > P(52k) > S(41k) > T(47years) > Q(42years)(37k) > R(29k) > U \text{ (15years)}$$



Answer: E

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